

Yug Dalwadi

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

Indian Institute of Technology, Jodhpur

2024–2028

B.Tech, Artificial Intelligence & Data Science

CGPA: 8.35

Expected Graduation: May 2028

EXPERIENCE

Inter IIT Tech Meet 14.0 — Real-Time Fraud Detection System [GitHub](#) | *Kafka, Pathway, River* Nov – Dec 2025

- Architected a production-grade streaming fraud detection system using **Kafka, Pathway, and River**, handling **70k+ txns/min** with **sub-40ms P95 latency**.
- Deployed online learning algorithms tuned using Optuna for real-time drift adaptation, reducing false positives by **15–20%**.
- Designed an automated retraining pipeline via **BigQuery, Celery, and W&B**, enabling **80% faster** model updates and zero-downtime deployment.

PROJECTS

Transformer Implementation [GitHub](#) | *PyTorch, W&B, Apache Arrow* Mar 2026

- Developed a **140M+ parameter** transformer with Attention from scratch, **Rotary Positional Embeddings (RoPE)**, causal masking, and tied embeddings, trained on the **8B+ token** OpenWebText dataset.
- Engineered an efficient data pipeline using **Apache Arrow** for sharded tokenization, significantly reducing I/O overhead during pre-training.
- Optimized single-GPU throughput via **Mixed Precision (FP16)** and **Gradient Accumulation**, integrating **Weights & Biases** for real-time loss and perplexity tracking.

Bahdanau Attention — Neural Machine Translation [GitHub](#) | *PyTorch, NumPy, spaCy* Jun 2025

- Built a **BiLSTM encoder-decoder** with Bahdanau Attention from scratch in PyTorch, trained on an English–French corpus of **135k sentence pairs** with custom tokenization and vocabulary construction using spaCy.
- Implemented a teacher forcing decay schedule, improving training convergence stability by **15–25%** over fixed teacher forcing, with smoother loss curves and better generalization at inference.

FileNest (Decentralized AI Search) [GitHub](#) | *Go, libp2p, Vector Embeddings* Jun – Nov 2025

- Built a P2P file discovery system utilizing **Kademlia-style routing** over go-libp2p, reducing lookup hops from $O(n)$ to $O(\log n)$ compared to naive linear search.
- Integrated **multimodal retrieval** (text + image) using **Nomic and FastVLM** embeddings, enabling users to semantically search for files using natural language queries or similar images.
- Reduced average query resolution time by **40%** through depth-aware routing and optimized vector indexing strategies.

SnapShot (AI Photo Management System) [GitHub](#) | *FastAPI, Pinecone, Docker* Jan – May 2025

- Created a microservices-based system for semantic image search using **FaceNet** embeddings and **Pinecone** vector database, supporting large-scale photo collections.
- Developed a CV pipeline for automated face-based clustering and albumization, achieving **sub-3s query latency**.

TECHNICAL SKILLS

- **Languages:** Python, C++, Go, SQL, C#, Lua
- **AI/ML:** PyTorch, Scikit-learn, River (Streaming ML), Transformer Architectures, Vector Search (Pinecone), LLM Pre-training
- **Systems & Tools:** Kafka, Docker, FastAPI, Django, PostgreSQL, Neo4j, Apache Arrow, Celery, Weights & Biases
- **Developer Tools:** Git, Linux (Bash), Microservices Architecture

COURSEWORK & ACTIVITIES

- **Relevant Coursework:** Pattern Recognition and Machine Learning (A-), Probability Statistics & Stochastic Processes (A-), Linear Algebra (A)
- **Clubs:** Core Team Member — Devlup (Open Source Club); Member — RAID (AI Club)
- **Mentorship:** WARP'26 — Mentored a project on depth-conditioned image generation via StyleGAN modification, organised by RAID (AI Club)